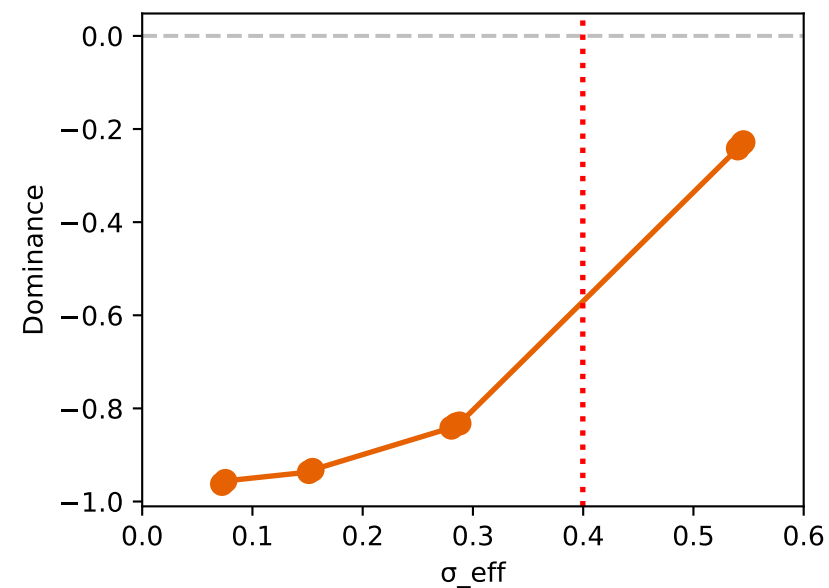
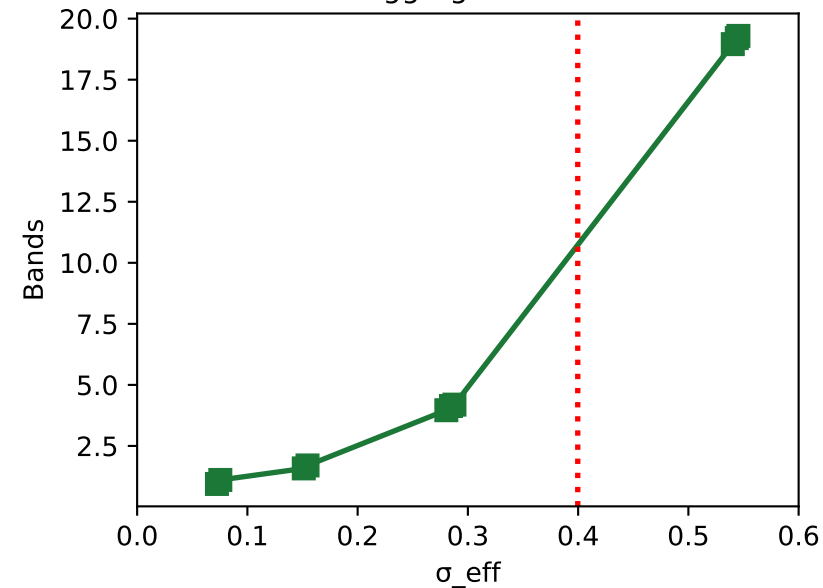


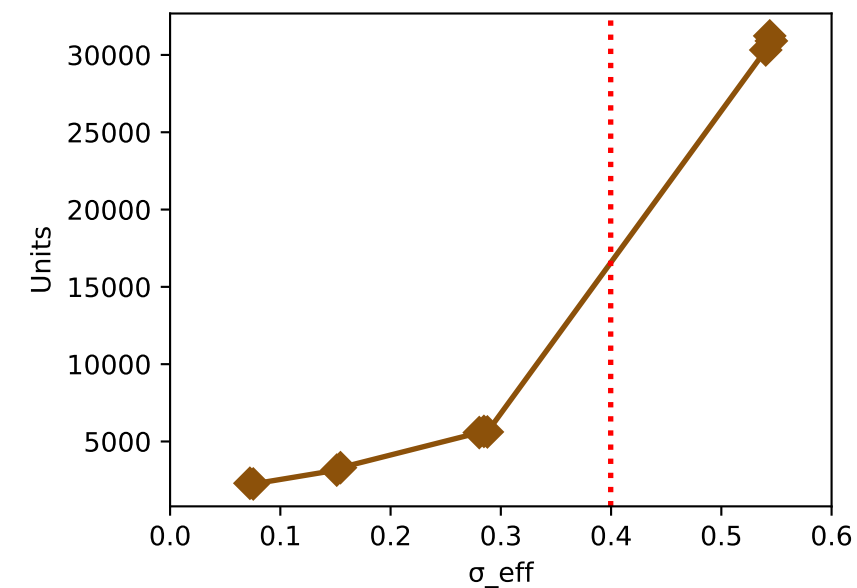
A. Phase Transition



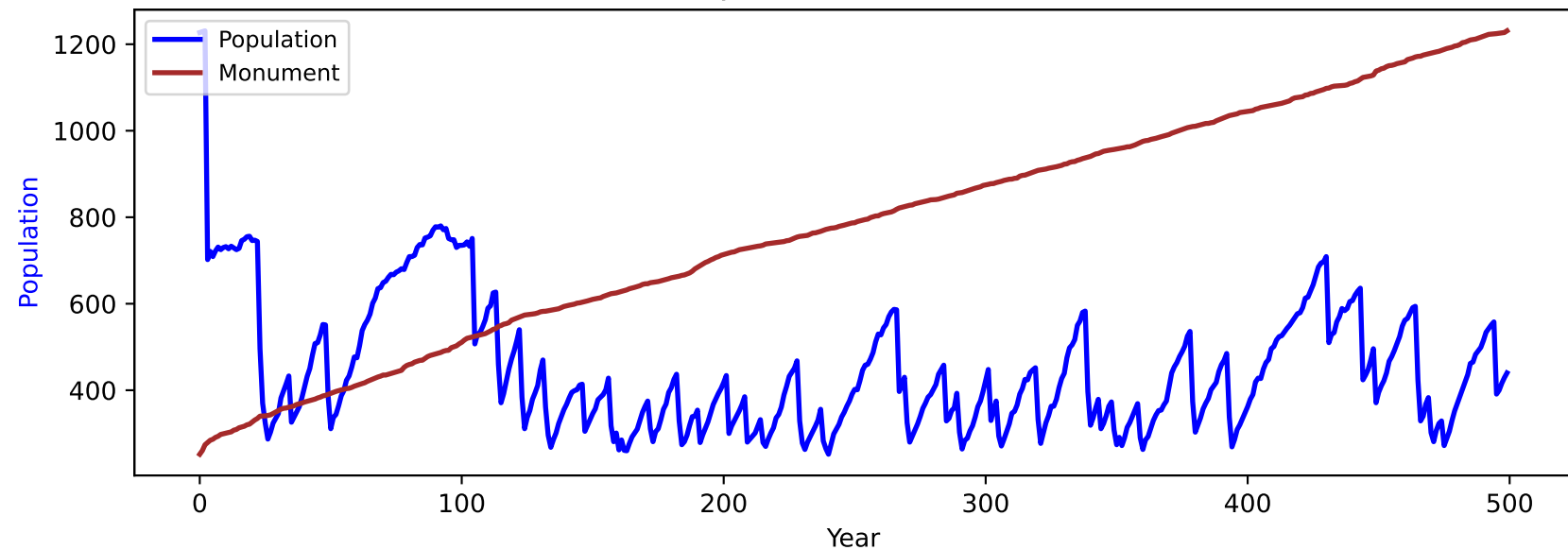
B. Aggregation Size



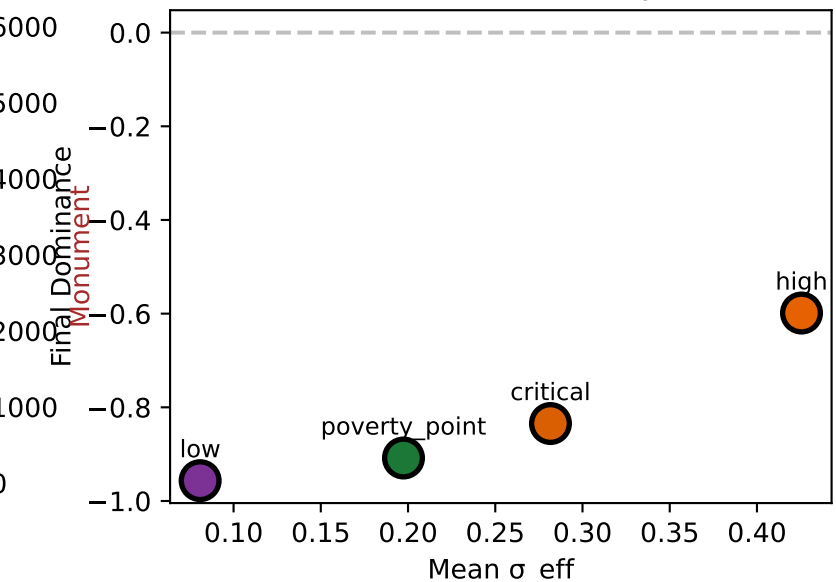
C. Monument Level



D. Poverty Point Scenario Time Series



E. Scenario Summary



KEY FINDINGS FROM INTEGRATED SIMULATION

- PHASE TRANSITION VALIDATED:** The model correctly predicts a transition from independent-dominated (dominance ≈ -0.95) to mixed strategy (dominance ≈ -0.24) as environmental uncertainty increases above $\sigma^* \approx 0.54$.
- AGGREGATION SIZE:** At $\sigma_{\text{eff}} > 0.54$, mean aggregation size jumps from ~ 4 bands to ~ 19 bands, approaching the theoretical optimal $n^* = 25$.
- MONUMENT SCALING:** Model monument units scale to archaeological data with factor of 142.6. At this scaling:
 - Model: 750,000 m³
 - Archaeological: $\sim 750,000$ m³
- SCENARIO COMPARISON:**
 - Low σ (0.08): 96% independent, minimal monument building
 - Poverty Point (0.20): 91% independent, moderate construction
 - High σ (0.43): 60% independent, sustained construction
 - Critical (0.28): 84% independent, near phase transition
- EXOTIC GOODS:** Total exotics range from 895 (low σ) to 5,265 (high σ), consistent with archaeological expectations.

CONCLUSION: The integrated simulation validates the core theoretical prediction: aggregation-based costly signaling emerges as an adaptive response to environmental uncertainty above a critical threshold.